WORKSHEET

Loan repayment problems

1 Bella borrows $18 000 at 5.6% p.a. compound interest, adjusted monthly. The loan is to be repaid in 9 years. Find:

a the monthly repayment

b the total amount repaid on the loan

c how much interest she pays on the loan.

2 Marco takes out a $120 000 business loan at an interest rate of 5.5% p.a. adjusted monthly. His monthly repayments are $857.

a How much will Marco still owe on the loan after 3 years?

b How long will it take him to repay the loan?

3 Kit takes out a loan for $80 000 which she repays in monthly instalments of $1230 over 8 years. Find the interest rate per month that Kit is charged for the loan.

4 Vita borrows $40 000 for a car that he intends to repay in 5 years. If the interest rate is 8.25% p.a., compounded monthly, find:

a the size of his monthly repayments

b the total amount he pays on the loan

c how much interest he pays.

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5 Mara borrows $190 000 at 6.75% p.a. compounded monthly. She can afford to make monthly repayments of $1475.

a How long will it take Mara to reduce the amount owing to below $100 000?

b How many repayments will Mara make in total for this loan?

c Mara makes all her repayments on the first day of the month. If her first repayment was on 1 January 2014, what will be the date of her last repayment?

6 Morgan borrows $24 000 to renovate his kitchen. He pays interest at a rate of 10.5% p.a., compounded quarterly. He would like to repay the loan in 7 years.

a What should be his quarterly repayment?

b How much interest will he pay on the loan?

7 Lucius borrows $400 000 to buy a house. He pays 4.5% interest per annum, compounded monthly, and he plans to repay the loan over 30 years.

a What will be his monthly repayment?

b How long will it take for Lucius to halve the amount he owes?

c How much interest will Lucius pay on this loan over the 30 years?

8 Sam borrows $35 000 to buy a caravan. Interest is charged at 11.4% p.a., adjusted monthly.

a If Sam was to repay the loan in 3 years, what would be his monthly repayment?

b How much would Sam pay in total for this loan?

c How much interest would he pay?

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Sam finds he can only afford to pay $750 per month for his loan.

d How long will it take him to repay the loan?

e How much more interest will he now pay rather than if he had repaid the loan in 3 years?

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Answers

1 **a** $212.56

b $22 956.48

c $4956.48

2 **a** $108 013.70

b 225 months or 18 years and 9 months

3 0.865%

4 **a** $815.85

b $48 951

c $8951

5 **a** 145 months or 12 years and 1 month

b 230 months or 19 years and 2 months

c 1 February 2033

6 **a** $1221.10

b $10 190.80

7 **a** $2026.74

b 237 months or 19 years and 9 months

c $329 626.40

8 **a** $1152.50

b $41 490

c $6490

d 62 months or 5 years and 2 months

e $5010

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